

Questions 1–20

Question 1

A healthcare organization wants to predict which patients are likely to miss follow-up appointments. During project planning, the AI team realizes that a simple SMS reminder system might solve the problem just as effectively.

What is the MOST responsible action?

- A. Continue building the AI model because AI is more innovative.
 - B. Build the AI model and compare it with SMS reminders after deployment.
 - C. Determine whether machine learning is actually necessary before development.
 - D. Use a more interpretable model.
-

Question 2

A bank uses a deep neural network for loan approvals. Regulators ask why one applicant was denied.

Which explainability technique would BEST provide an explanation for this individual prediction?

- A. Demographic Parity
 - B. SHAP
 - C. Data Normalization
 - D. PCA
-

Question 3

An organization discovers that applicants from rural regions are rejected significantly more often.

The model does not use geographic location directly, but uses internet speed as an input.

What is the MOST likely explanation?

- A. Overfitting
 - B. Label leakage
 - C. Proxy feature bias
 - D. Hyperparameter optimization
-

Question 4

A development team balances the training dataset by oversampling underrepresented applicants before model training.

This is an example of:

- A. Post-processing mitigation
 - B. In-processing mitigation
 - C. Pre-processing mitigation
 - D. Model monitoring
-

Question 5

Which Responsible AI principle is MOST directly supported by documenting every model version, training dataset, and evaluation metric?

- A. Privacy
 - B. Accountability
 - C. Beneficence
 - D. Safety
-

Question 6

A hospital AI predicts patient deterioration.

Doctors always make the final treatment decision after reviewing the AI recommendation.

This BEST represents:

- A. Fairness
 - B. Non-maleficence (Beneficence)
 - C. Statistical parity
 - D. Model calibration
-

Question 7

Which stakeholder would MOST benefit from detailed error analysis, training procedures, and code traceability?

- A. Customers
 - B. Board of Directors
 - C. Data Scientists
 - D. Business Executives
-

Question 8

A highly accurate skin cancer model actually learns to recognize rulers appearing in medical images rather than cancer characteristics.

This primarily demonstrates why:

- A. More training data always improves fairness.
 - B. Explainability helps detect spurious correlations.
 - C. Neural networks should never be used.
 - D. SHAP guarantees model fairness.
-

Question 9

Which fairness approach attempts to ensure that similar individuals receive similar outcomes?

- A. Group Fairness
 - B. Demographic Parity
 - C. Individual Fairness
 - D. Predictive Parity
-

Question 10

A company deploys an AI fraud detection model.

Six months later, customer demographics change substantially.

What should the organization have already established?

- A. Larger neural network architecture
 - B. Model retraining criteria
 - C. Additional SHAP plots
 - D. New optimization algorithm
-

Question 11

Which explainability method approximates a complex model locally using a simpler surrogate model?

- A. SHAP
 - B. LIME
 - C. Decision Tree
 - D. Logistic Regression
-

Question 12

A regulator wants assurance that a lending model performs fairly across different ethnic groups.

Which metric would MOST likely be evaluated?

- A. Learning rate
 - B. Accuracy only
 - C. Demographic Parity
 - D. Feature Scaling
-

Question 13

Which documentation practice is MOST important during data collection?

- A. Recording GPU utilization
 - B. Informing participants how data will be used and protected
 - C. Choosing the best optimizer
 - D. Hyperparameter tuning
-

Question 14

A model card should primarily help stakeholders understand:

- A. Only the source code
 - B. Intended use, limitations, and performance
 - C. GPU specifications
 - D. Cloud deployment architecture
-

Question 15

Which statement BEST describes SHAP?

- A. It only explains neural networks.
 - B. It is based on cooperative game theory.
 - C. It modifies model weights.
 - D. It is only useful for local explanations.
-

Question 16

A model achieves identical overall accuracy for two demographic groups but has a much lower true positive rate for one group.

Which fairness metric would BEST identify this issue?

- A. Equal Opportunity
 - B. Model Complexity
 - C. Precision
 - D. MAE
-

Question 17

A company purchases a proprietary AI model and cannot retrain it.

Which bias mitigation approach is MOST practical?

- A. Pre-processing
 - B. In-processing
 - C. Post-processing
 - D. Feature Engineering
-

Question 18

Board members primarily require:

- A. Source code
 - B. Hyperparameter tuning logs
 - C. High-level AI risks, governance, and strategy
 - D. Confusion matrices
-

Question 19

Which statement about transparency is MOST accurate?

- A. Transparency only benefits developers.
 - B. Transparency improves auditability and stakeholder trust.
 - C. Transparency eliminates bias.
 - D. Transparency always improves accuracy.
-

Question 20

Which explainability approach is generally preferred when predictive performance requirements can be satisfied using a simple model?

- A. Deep Neural Networks with SHAP
 - B. Inherently interpretable models
 - C. Ensemble methods
 - D. Black-box optimization
-

Answer Key with Explanations

Q Answer Explanation

- | | | |
|----|----------|---|
| 1 | C | The pre-development phase first asks whether AI is needed at all before investing in ML development. |
| 2 | B | SHAP provides feature-level explanations for individual predictions and is widely used in regulated environments. |
| 3 | C | Internet speed indirectly represents geography, making it a proxy feature that can introduce bias. |
| 4 | C | Oversampling and re-weighting occur before training, making them pre-processing techniques. |
| 5 | B | Documentation creates traceability, auditability, and responsibility, supporting accountability. |
| 6 | B | Keeping humans responsible for major decisions aligns with beneficence (non-maleficence). |
| 7 | C | Technical stakeholders require detailed model internals, performance metrics, and traceability. |
| 8 | B | Explainability uncovers shortcut learning and spurious correlations, like the ruler example discussed in the study guide. |
| 9 | C | Individual fairness focuses on treating similar individuals similarly. |
| 10 | B | Organizations should establish monitoring and clear retraining criteria before deployment. |
| 11 | B | LIME builds a simple local surrogate model around one prediction. |
| 12 | C | Demographic parity evaluates whether positive outcomes are balanced across protected groups. |
| 13 | B | Transparency begins with informing participants how data is collected, stored, and protected. |

Q Answer Explanation

- 14 **B** Model cards document intended use, limitations, assumptions, and performance characteristics.
- 15 **B** SHAP is mathematically grounded in cooperative game theory.
- 16 **A** Equal Opportunity focuses on equal true positive rates across groups.
- 17 **C** When retraining is impossible (e.g., a black-box model), post-processing is the practical mitigation method.
- 18 **C** Boards require strategic information on governance, risks, accountability, and AI oversight.
- 19 **B** Transparency improves trust, auditability, and accountability but does not automatically eliminate bias.
- 20 **B** If a simple interpretable model meets performance needs, it is generally preferred over complex black-box models.
-

These questions were designed to mirror the style of the official sample—focusing on realistic scenarios, subtle distinctions between concepts (e.g., fairness metrics, lifecycle phases, SHAP vs. LIME), and application rather than rote recall—while aligning with the Module 4 learning objectives and content.

Questions 21–40

Question 21

An insurance company develops an AI model to predict fraudulent claims. During model evaluation, engineers discover that claimants over age 65 experience significantly more false positives than younger applicants.

Which fairness metric would MOST directly highlight this issue?

- A. Demographic Parity
 - B. Equalized Odds
 - C. Model Accuracy
 - D. Mean Absolute Error
-

Question 22

Which statement BEST distinguishes interpretability from explainability?

- A. Interpretability and explainability are identical concepts.
 - B. Interpretability refers to understanding a model by design, while explainability provides justifications for model decisions.
 - C. Explainability applies only to neural networks.
 - D. Interpretability only applies after deployment.
-

Question 23

A financial institution must decide between a decision tree with 92% accuracy and a deep neural network with 95% accuracy for regulatory lending decisions.

Which consideration should MOST influence the final decision?

- A. Select the model with the highest accuracy regardless of context.
 - B. Balance predictive performance with interpretability and regulatory requirements.
 - C. Always choose the simpler model.
 - D. Use whichever model trains faster.
-

Question 24

Which of the following is an example of **global** model explanation?

- A. Explaining why a single customer was denied a mortgage.
 - B. Identifying which features influence predictions across the entire dataset.
 - C. Explaining one image classification result.
 - D. Providing recourse for one rejected applicant.
-

Question 25

Why is "fairness through unawareness" often insufficient?

- A. Sensitive attributes are always unavailable.
 - B. Proxy features may still indirectly reveal sensitive information.
 - C. It decreases model accuracy.
 - D. It requires too much computational power.
-

Question 26

An ML engineer incorporates demographic parity directly into the model's loss function.

This represents:

- A. Pre-processing
 - B. In-processing
 - C. Post-processing
 - D. Model monitoring
-

Question 27

A company discovers that applicants with identical qualifications receive different hiring recommendations because they attended universities in different postal codes that correlate with ethnicity.

Which concept BEST explains this issue?

- A. Feature scaling
 - B. Proxy discrimination
 - C. Cross-validation
 - D. Data normalization
-

Question 28

Which explainability technique is generally **more computationally expensive**?

- A. LIME
 - B. Decision Trees
 - C. SHAP
 - D. Logistic Regression
-

Question 29

Why are counterfactual explanations valuable?

- A. They improve model accuracy.
 - B. They identify minimal input changes that would change a prediction.
 - C. They eliminate bias completely.
 - D. They replace SHAP values.
-

Question 30

A regulator asks an organization to demonstrate that customers with the same predicted repayment probability actually repay loans at similar rates across demographic groups.

Which fairness metric is MOST relevant?

- A. Equal Opportunity
 - B. Predictive Parity (Calibration)
 - C. Demographic Parity
 - D. Recall
-

Question 31

A company notices declining model performance after deployment due to changes in customer behavior.

This phenomenon is MOST closely associated with:

- A. Data drift
 - B. Gradient descent
 - C. Hyperparameter tuning
 - D. Model compression
-

Question 32

Which Responsible AI principle is MOST directly concerned with minimizing harm while maximizing societal benefit?

- A. Fairness
 - B. Accountability
 - C. Beneficence
 - D. Transparency
-

Question 33

A data scientist uses SHAP values to identify that "credit score" contributes +0.45 toward loan approval while "debt ratio" contributes -0.62.

This illustrates that SHAP:

- A. Produces quantitative feature contributions.
- B. Retrains the model.
- C. Measures fairness.

D. Detects adversarial attacks.

Question 34

Which stakeholder would MOST likely require model cards, fairness audits, and compliance evidence?

- A. Customers
 - B. Regulators
 - C. Marketing teams
 - D. Software developers
-

Question 35

During exploratory data analysis (EDA), analysts discover severe class imbalance.

Why is this discovery important?

- A. It determines deployment architecture.
 - B. It may introduce biased learning if not addressed.
 - C. It guarantees overfitting.
 - D. It improves explainability.
-

Question 36

A company decides to lower decision thresholds for a disadvantaged population after model training.

This is an example of:

- A. Re-weighting
 - B. Threshold Adjustment
 - C. Adversarial Debiasing
 - D. Fairness Through Awareness
-

Question 37

Which statement BEST describes LIME?

- A. It explains every prediction using cooperative game theory.
- B. It approximates complex model behavior locally using simpler models.
- C. It only works for decision trees.

D. It guarantees globally consistent explanations.

Question 38

Why should organizations document known failure modes?

- A. To improve GPU performance.
 - B. To enable stakeholders to make risk-aware decisions.
 - C. To increase training speed.
 - D. To eliminate model uncertainty.
-

Question 39

A customer asks, "What could I improve so that my loan application would likely be approved?"

Which benefit of model understanding is MOST applicable?

- A. Monitoring
 - B. Recourse
 - C. Deployment
 - D. Calibration
-

Question 40

An AI team monitors fairness metrics monthly after deployment and has predefined conditions for retraining the model.

This BEST demonstrates:

- A. Continuous Responsible AI governance.
 - B. Explainability.
 - C. Hyperparameter optimization.
 - D. Model compression.
-

Answer Key with Explanations

Q Answer Explanation

- 21 **B** Equalized Odds evaluates both True Positive Rate (TPR) and False Positive Rate (FPR). Since older applicants experience more false positives, Equalized Odds would detect this disparity.

Q Answer Explanation

- 22 **B** Interpretability refers to models that are understandable by design (e.g., decision trees), whereas explainability provides reasons for predictions, especially for complex models.
- 23 **B** TAISE emphasizes balancing predictive performance with interpretability, regulatory expectations, and stakeholder trust—not maximizing accuracy alone.
- 24 **B** Global explanations identify overall feature importance and behavior across the model, unlike local explanations that explain individual predictions.
- 25 **B** Removing protected attributes does not eliminate bias because proxy variables can still encode sensitive information indirectly.
- 26 **B** Adding fairness constraints into the loss function modifies the learning algorithm during training, making it an in-processing mitigation technique.
- 27 **B** Postal codes indirectly reveal ethnicity, making them proxy features that can introduce discrimination even when ethnicity is excluded.
- 28 **C** SHAP evaluates feature contributions across many feature combinations, making it computationally more expensive than LIME.
- 29 **B** Counterfactual explanations tell users the smallest changes needed to receive a different prediction, supporting transparency and recourse.
- 30 **B** Predictive Parity (Calibration) ensures that predicted probabilities correspond to actual outcomes equally across demographic groups.
- 31 **A** Changes in production data distributions over time are known as data drift and require monitoring and possible retraining.
- 32 **C** Beneficence (Non-maleficence) focuses on promoting human well-being while minimizing potential harm.
- 33 **A** SHAP assigns numerical contributions to each feature, allowing users to quantify how much each feature influenced a prediction.
- 34 **B** Regulators and auditors require structured documentation, fairness assessments, and compliance evidence to evaluate AI governance.
- 35 **B** Class imbalance can bias model learning and should be addressed before training through techniques such as re-sampling or re-weighting.
- 36 **B** Adjusting decision thresholds after model training is a post-processing bias mitigation technique known as threshold adjustment.
- 37 **B** LIME explains individual predictions by fitting a simple surrogate model locally around the prediction being analyzed.

Q Answer Explanation

- 38 B Documenting failure modes enables stakeholders to understand limitations and make informed, risk-aware decisions.
- 39 B Recourse provides users with actionable guidance about what factors they could change to obtain a different prediction in the future.
- 40 A Continuous monitoring, fairness evaluation, and predefined retraining criteria are core elements of Responsible AI governance throughout the AI lifecycle.

Questions 41–60

Difficulty: High (Designed to closely resemble the reasoning and application style of the TAISE certification exam.)

Question 41

A financial institution develops an AI model for mortgage approvals. During validation, auditors discover that applicants from one demographic group receive significantly more loan approvals than equally qualified applicants from another group.

The model has excellent overall accuracy.

What is the MOST appropriate conclusion?

- A. The model is acceptable because accuracy is high.
 - B. Accuracy alone is insufficient; fairness metrics should also be evaluated.
 - C. The model should immediately be discarded.
 - D. More training data always eliminates fairness issues.
-

Question 42

An organization wants to explain why one particular patient was classified as high risk.

Which explainability approach is MOST appropriate?

- A. Global feature importance
 - B. Local explanation
 - C. Confusion Matrix
 - D. Demographic Parity
-

Question 43

Why are confusion matrices useful when evaluating fairness?

- A. They calculate SHAP values.
 - B. They allow comparison of classification errors across demographic groups.
 - C. They improve model accuracy.
 - D. They reduce computational complexity.
-

Question 44

A retailer notices that its recommendation engine consistently favors products purchased by historically dominant customer groups.

Which Responsible AI principle is MOST directly affected?

- A. Privacy
 - B. Fairness
 - C. Accountability
 - D. Security
-

Question 45

An AI team documents every preprocessing step, including missing value treatment, feature encoding, and bias corrections.

The PRIMARY benefit is:

- A. Faster model training.
 - B. Improved transparency and reproducibility.
 - C. Reduced computational cost.
 - D. Better cloud performance.
-

Question 46

A regulator asks whether an AI model behaves consistently across different demographic groups.

Which type of explanation is MOST useful?

- A. Local explanation
 - B. Global explanation
 - C. Integrated Gradients
 - D. Saliency Map
-

Question 47

A healthcare model predicts disease risk.

Which situation BEST illustrates a **spurious correlation**?

- A. Blood pressure strongly predicts heart disease.
 - B. Patient age predicts retirement.
 - C. Images containing hospital logos are classified as cancerous because most cancer images came from one hospital.
 - D. Smoking predicts lung disease.
-

Question 48

Which statement BEST describes accountability?

- A. Ensuring every prediction is accurate.
 - B. Establishing governance mechanisms to oversee AI and remediate issues.
 - C. Eliminating human involvement.
 - D. Maximizing transparency regardless of privacy.
-

Question 49

An AI development team continuously reviews customer complaints after deployment.

Which Responsible AI principle is MOST strengthened?

- A. Accountability
 - B. Interpretability
 - C. Feature Engineering
 - D. Hyperparameter Optimization
-

Question 50

A deep learning model achieves superior performance compared with a logistic regression model.

Which statement BEST aligns with the Module 4 guidance?

- A. Always deploy the deep learning model.
- B. Always prefer logistic regression.
- C. Consider whether additional predictive performance justifies reduced interpretability.
- D. Ignore explainability if performance is higher.

Question 51

Which explainability method is MOST appropriate for explaining image classification models built using Convolutional Neural Networks (CNNs)?

- A. Grad-CAM
- B. PCA
- C. Logistic Regression
- D. ROC Curve

Question 52

Which fairness metric focuses specifically on ensuring equal **True Positive Rates** across demographic groups?

- A. Equal Opportunity
- B. Demographic Parity
- C. Predictive Parity
- D. Statistical Significance

Question 53

A company excludes gender from training data but still observes gender discrimination because shopping behavior indirectly predicts gender.

This illustrates:

- A. Underfitting
- B. Fairness through Awareness
- C. Proxy Features
- D. Cross Validation

Question 54

A model explanation identifies that increasing annual income by \$8,000 would likely change a loan denial into an approval.

This is BEST described as:

- A. SHAP Explanation
- B. Counterfactual Explanation

C. Feature Scaling

D. Calibration

Question 55

During deployment, business executives argue that releasing the model immediately will maximize quarterly revenue despite fairness concerns.

According to Responsible AI principles, leadership should:

- A. Ignore fairness because business objectives take priority.
 - B. Delay deployment until every possible risk is eliminated.
 - C. Evaluate trade-offs using documented risks before deciding.
 - D. Deploy first and evaluate fairness later.
-

Question 56

Which stakeholder is MOST interested in understanding:

- AI governance
 - Organizational risks
 - Strategic objectives
 - Regulatory exposure
- A. Data Scientist
 - B. End User
 - C. Board of Directors
 - D. Software Developer
-

Question 57

Which statement BEST describes **model understanding**?

- A. Measuring computational efficiency.
 - B. Understanding why a model reaches particular decisions.
 - C. Reducing training time.
 - D. Selecting cloud infrastructure.
-

Question 58

An organization retrains its fraud detection model every six months regardless of performance.

Which practice would better align with Responsible AI?

- A. Continue retraining on a fixed schedule only.
 - B. Retrain only when predefined monitoring criteria indicate performance degradation.
 - C. Never retrain.
 - D. Retrain after customer complaints only.
-

Question 59

Which statement BEST explains why transparency may sometimes conflict with privacy?

- A. Transparency always reduces privacy.
 - B. Explaining model behavior may reveal sensitive personal information.
 - C. Privacy eliminates explainability.
 - D. Transparent systems cannot comply with regulations.
-

Question 60

Which sequence BEST represents the AI Model Development Lifecycle?

- A. Development → Deployment → Monitoring → Data Collection
 - B. Data Collection → Deployment → Evaluation → Monitoring
 - C. Pre-development → Development → Deployment → Post-deployment
 - D. Development → Evaluation → Monitoring → Pre-development
-

Answer Key with Explanations

Q Answer Explanation

- | | |
|------|---|
| 41 B | High predictive accuracy does not guarantee fairness. Responsible AI requires evaluating fairness metrics alongside traditional performance measures. |
| 42 B | A local explanation answers why one specific prediction was made, making it ideal for individual cases. |
| 43 B | Confusion matrices enable comparison of TPR, FPR, precision, and recall across demographic groups, supporting fairness assessment. |
| 44 B | Unequal treatment across customer groups is fundamentally a fairness issue. |

Q Answer Explanation

- 45 B Detailed preprocessing documentation improves transparency, reproducibility, debugging, and auditing.
- 46 B Global explanations reveal overall model behavior across populations, which regulators often need to assess systemic fairness.
- 47 C The model learned an irrelevant shortcut (hospital logo) instead of genuine medical features, a classic spurious correlation.
- 48 B Accountability establishes governance, oversight, auditing, monitoring, and remediation processes—not merely assigning blame.
- 49 A Monitoring customer complaints creates feedback mechanisms that strengthen accountability and continuous governance.
- 50 C The study guide emphasizes balancing predictive performance against interpretability rather than maximizing either one in isolation.
- 51 A Grad-CAM highlights image regions that most influenced CNN predictions, making it suitable for image explainability.
- 52 A Equal Opportunity requires equal True Positive Rates across protected groups.
- 53 C Shopping behavior indirectly reveals gender, making it a proxy feature that can perpetuate discrimination.
- 54 B Counterfactual explanations identify minimal changes needed to obtain a different prediction, providing actionable recourse.
- 55 C Deployment decisions should balance business value with documented fairness risks and potential impacts on affected individuals.
- 56 C Board members focus on governance, organizational risk, compliance, accountability, and strategic AI oversight.
- 57 B Model understanding focuses on how and why an AI model arrives at its predictions, supporting trust and validation.
- 58 B Responsible AI recommends continuous monitoring with predefined retraining criteria rather than relying solely on fixed schedules.
- 59 B Detailed explanations may inadvertently expose sensitive personal information or enable misuse, creating tension between transparency and privacy.
- 60 C The lifecycle proceeds through **Pre-development → Development → Deployment → Post-deployment**, with responsible AI practices integrated throughout.

Questions 61–80

Difficulty: Advanced (Designed to mirror the reasoning, scenario analysis, and subtle distractors expected in the TAISE certification exam.)

Question 61

A bank develops a highly accurate credit approval model. During testing, SHAP analysis reveals that applicants living in certain ZIP codes consistently receive lower approval scores, even though ZIP code was not included as a feature.

What is the MOST likely explanation?

- A. Model overfitting
 - B. Proxy feature bias
 - C. Hyperparameter instability
 - D. Data normalization errors
-

Question 62

An organization wants to increase transparency without sacrificing the predictive performance of its deep neural network.

Which approach is MOST appropriate?

- A. Replace the neural network with logistic regression regardless of performance.
 - B. Apply post-hoc explainability techniques such as SHAP or LIME.
 - C. Remove all hidden layers.
 - D. Train only on balanced datasets.
-

Question 63

During model development, engineers discover that female applicants are underrepresented in the training dataset.

Which action should be taken FIRST?

- A. Deploy the model and monitor results.
 - B. Investigate dataset representativeness and apply bias mitigation if necessary.
 - C. Increase model complexity.
 - D. Lower the prediction threshold.
-

Question 64

An AI model predicts hospital readmission.

Which stakeholder would MOST benefit from a counterfactual explanation?

- A. Data Scientist
 - B. Patient
 - C. Internal Auditor
 - D. Board Member
-

Question 65

A company wants to explain how income affects loan approval across the entire customer population.

Which explainability method is MOST appropriate?

- A. Local explanation
 - B. Global explanation
 - C. Counterfactual explanation
 - D. Recourse explanation
-

Question 66

Which of the following BEST illustrates **accountability**?

- A. Publishing model accuracy.
 - B. Assigning clear ownership for responding to model failures.
 - C. Using explainable algorithms.
 - D. Encrypting personal data.
-

Question 67

A regulator asks whether an AI hiring model rejects qualified candidates equally across demographic groups.

Which fairness metric BEST addresses this concern?

- A. Equal Opportunity
 - B. Learning Rate
 - C. SHAP Value
 - D. Accuracy
-

Question 68

Which statement BEST describes **Demographic Parity**?

- A. Every individual receives the same prediction.
 - B. Positive prediction rates should be similar across protected groups.
 - C. True positive rates should always equal false positive rates.
 - D. Similar individuals should always receive identical outcomes.
-

Question 69

A company trains an AI model using historical hiring data that reflects decades of biased hiring decisions.

What is the PRIMARY risk?

- A. Data drift
 - B. Historical bias being learned by the model
 - C. Model compression
 - D. Gradient explosion
-

Question 70

Which explanation technique is generally **model-agnostic**?

- A. LIME
 - B. Decision Trees
 - C. Logistic Regression
 - D. Linear Regression
-

Question 71

An AI model correctly predicts repayment probability but consistently provides poor explanations to customers.

Which Responsible AI principle is MOST affected?

- A. Explainability
 - B. Availability
 - C. Performance
 - D. Scalability
-

Question 72

A fraud detection model performs equally well during testing but deteriorates significantly six months after deployment.

What is the MOST likely cause?

- A. Hyperparameter tuning
 - B. Data drift
 - C. Over-regularization
 - D. Gradient clipping
-

Question 73

Which document is MOST useful for communicating a model's intended use, limitations, ethical considerations, and performance?

- A. Source Code Repository
 - B. Model Card
 - C. Confusion Matrix
 - D. Feature Store
-

Question 74

An organization uses **Adversarial Debiasing** during model training.

This technique belongs to which category?

- A. Pre-processing
 - B. In-processing
 - C. Post-processing
 - D. Monitoring
-

Question 75

Which stakeholder is MOST interested in understanding:

- fairness audits
- compliance evidence
- governance documentation
- model risk assessments

- A. End User

- B. Regulators
 - C. Product Manager
 - D. Customer Support
-

Question 76

An AI model predicts that an applicant has a 75% chance of loan repayment.

Predictive Parity requires that:

- A. Every demographic group has identical approval rates.
 - B. A 75% prediction should represent approximately the same real repayment probability across demographic groups.
 - C. False positive rates must be equal.
 - D. Everyone should receive identical scores.
-

Question 77

A development team discovers that changing a customer's debt-to-income ratio from 52% to 39% would change a loan rejection into an approval.

This represents:

- A. SHAP Analysis
 - B. Counterfactual Explanation
 - C. Equalized Odds
 - D. Demographic Parity
-

Question 78

Which activity is MOST appropriate during the **post-deployment** phase?

- A. Hyperparameter tuning
 - B. Exploratory Data Analysis
 - C. Continuous monitoring for performance degradation and fairness
 - D. Initial stakeholder interviews
-

Question 79

Why should organizations communicate **what an AI model cannot do**?

- A. To reduce infrastructure costs.

- B. To enable stakeholders to make informed, risk-aware decisions.
 - C. To improve training speed.
 - D. To increase model complexity.
-

Question 80

An organization has limited computational resources but still requires local explanations for individual predictions.

Which explainability technique would generally be the MOST practical?

- A. SHAP
 - B. LIME
 - C. Grad-CAM
 - D. Integrated Gradients
-

Answer Key with Explanations

Q Answer Explanation

- | | |
|------|--|
| 61 B | Even without ZIP code, correlated variables may act as proxy features , introducing indirect discrimination. |
| 62 B | Post-hoc methods like SHAP and LIME improve explainability while retaining the performance of complex models. |
| 63 B | Dataset representativeness should be evaluated before training continues, as biased data often produces biased models. |
| 64 B | Counterfactual explanations help patients understand what factors would change an individual prediction, supporting recourse. |
| 65 B | Global explanations identify feature importance and behavior across the entire population rather than a single prediction. |
| 66 B | Accountability requires defined governance, ownership, oversight, and remediation processes when AI issues occur. |
| 67 A | Equal Opportunity compares True Positive Rates among qualified applicants across protected groups. |
| 68 B | Demographic Parity requires similar positive prediction rates across protected groups, regardless of ground-truth outcome distributions. |
| 69 B | Historical bias embedded in training data can be learned and perpetuated by the model if not addressed. |

Q Answer Explanation

- 70 A LIME is a **model-agnostic** explainability method that can explain many different types of machine learning models.
- 71 A Accurate predictions alone are insufficient; users also need understandable explanations, making explainability the affected principle.
- 72 B Production data often changes over time (data drift), degrading model performance and requiring monitoring and possible retraining.
- 73 B Model Cards summarize intended use, limitations, performance, ethical considerations, and deployment context.
- 74 B Adversarial Debiasing modifies the learning process during training, making it an **in-processing** bias mitigation technique.
- 75 B Regulators and auditors require evidence of governance, fairness, compliance, and documented risk management.
- 76 B Predictive Parity (Calibration) requires predicted probabilities to have the same meaning across demographic groups.
- 77 B Counterfactual explanations identify the minimal input changes that would alter the prediction.
- 78 C Post-deployment focuses on continuous monitoring, detecting drift, measuring fairness, and determining when retraining is needed.
- 79 B Clearly communicating limitations and failure modes helps stakeholders make better risk-aware decisions and builds trust.
- 80 B LIME generally provides local explanations with lower computational cost than SHAP, making it practical when resources are constrained.

Question 81

A healthcare organization develops an AI model to prioritize patients for specialist appointments. The model demonstrates excellent overall accuracy, but post-deployment monitoring reveals that patients from rural communities consistently experience longer wait times because the model relies heavily on travel history.

What is the MOST appropriate initial response?

- A. Replace the model immediately.
- B. Investigate whether travel history acts as a proxy feature contributing to unfair outcomes.
- C. Increase the model's complexity.

D. Ignore the issue because the model is accurate.

Question 82

A financial institution chooses a complex ensemble model because it significantly outperforms simpler models. Regulators require explanations for every individual loan decision.

Which approach BEST balances these competing requirements?

- A. Replace the ensemble with logistic regression.
 - B. Apply SHAP explanations to the deployed model.
 - C. Remove important predictive variables.
 - D. Explain only rejected applications.
-

Question 83

Which scenario BEST demonstrates the principle of **Beneficence (Non-maleficence)**?

- A. A hospital AI autonomously performs surgery without physician oversight.
 - B. An AI system recommends treatment options, while physicians retain final authority.
 - C. An AI hiring system automatically rejects applicants.
 - D. An AI trading algorithm executes trades without human supervision.
-

Question 84

During model validation, engineers observe identical demographic parity across groups but significantly different False Positive Rates.

Which statement is MOST accurate?

- A. The model satisfies every fairness definition.
 - B. Different fairness metrics evaluate different aspects of fairness.
 - C. Demographic parity automatically guarantees equalized odds.
 - D. The model should only be evaluated using accuracy.
-

Question 85

An organization discovers that its customer service chatbot performs well during testing but struggles with regional dialects after deployment.

Which lifecycle activity should have detected this risk earlier?

- A. Hyperparameter tuning

- B. Representative data collection and stakeholder analysis during pre-development
 - C. GPU optimization
 - D. Model compression
-

Question 86

Which stakeholder is MOST likely to require documentation regarding:

- Model version history
- Training procedures
- Error analysis
- Feature engineering decisions

- A. Board Members
 - B. Technical Stakeholders
 - C. End Users
 - D. Marketing Teams
-

Question 87

A model card states that a medical AI should **not** be used for pediatric patients because it was trained only on adult data.

This information primarily communicates:

- A. Feature importance
 - B. Model limitations and intended use
 - C. Hyperparameters
 - D. Data encryption methods
-

Question 88

A model predicts employee performance.

Engineers discover that attendance records are strongly correlated with disability accommodations.

Why is this concerning?

- A. Attendance always improves accuracy.
- B. Attendance may function as a proxy for a protected attribute.
- C. Attendance reduces explainability.

D. Attendance causes overfitting.

Question 89

Which explainability method is BEST suited for understanding which pixels influenced an image classification produced by a CNN?

- A. LIME
 - B. Grad-CAM
 - C. Logistic Regression
 - D. Demographic Parity
-

Question 90

An organization establishes the following process:

- Monitor fairness monthly
- Monitor drift continuously
- Retrain only when predefined thresholds are exceeded

This BEST demonstrates:

- A. Continuous Responsible AI governance.
 - B. Explainability.
 - C. Fairness through unawareness.
 - D. Data preprocessing.
-

Question 91

A customer asks:

"What specifically caused my mortgage application to be denied?"

Which explainability capability BEST answers this question?

- A. Global feature importance
 - B. Local explanation
 - C. Demographic parity
 - D. Model monitoring
-

Question 92

Why are **Model Cards** valuable for regulators?

- A. They replace fairness audits.
 - B. They provide structured documentation regarding intended use, performance, limitations, and risks.
 - C. They improve model accuracy.
 - D. They automatically detect bias.
-

Question 93

A company uses SHAP to explain every prediction.

Which statement is TRUE?

- A. SHAP explanations guarantee fairness.
 - B. SHAP explains feature contributions but does not determine whether outcomes are fair.
 - C. SHAP eliminates proxy features.
 - D. SHAP replaces model documentation.
-

Question 94

Which situation MOST likely requires **post-processing** bias mitigation?

- A. The training dataset has class imbalance.
 - B. The organization cannot retrain a proprietary vendor model.
 - C. Engineers are selecting model features.
 - D. Dataset collection has not yet begun.
-

Question 95

A bank notices that applicants with identical repayment probabilities receive different approval decisions because different thresholds are applied across business units.

What governance weakness is MOST evident?

- A. Lack of monitoring
 - B. Inconsistent deployment and decision governance
 - C. Missing training data
 - D. Poor feature engineering
-

Question 96

Which statement BEST explains why explainability supports debugging?

- A. It always improves accuracy.
 - B. It reveals the features and reasoning driving predictions, allowing engineers to detect unintended behavior.
 - C. It reduces training costs.
 - D. It automatically removes bias.
-

Question 97

A model satisfies Equal Opportunity but not Predictive Parity.

What is the MOST reasonable interpretation?

- A. One fairness metric does not guarantee satisfaction of another.
 - B. The model is unbiased.
 - C. Equal Opportunity is always superior.
 - D. Predictive Parity is unnecessary.
-

Question 98

During deployment approval, executives observe excellent financial performance projections but documented fairness concerns affecting a protected group.

According to Responsible AI principles, leadership should:

- A. Ignore fairness because projected revenue is high.
 - B. Evaluate documented trade-offs and societal impacts before approving deployment.
 - C. Deploy immediately and address fairness after customer complaints.
 - D. Replace the model with manual processing.
-

Question 99

Which activity BEST demonstrates transparency throughout the AI lifecycle?

- A. Publishing only model accuracy.
 - B. Maintaining documentation across data collection, preprocessing, development, deployment, monitoring, and version history.
 - C. Encrypting training data.
 - D. Using explainable algorithms only.
-

Question 100

Which statement BEST summarizes the primary goal of Module 4?

- A. Build the most accurate AI model possible.
 - B. Replace human decision makers with AI.
 - C. Develop AI systems that are ethical, transparent, explainable, fair, accountable, and effectively communicated to stakeholders throughout the lifecycle.
 - D. Maximize automation while minimizing documentation.
-

Answer Key with Explanations

Q Answer Explanation

- | | | |
|----|----------|---|
| 81 | B | Before replacing the model, the team should investigate whether travel history is acting as a proxy feature that creates unfair outcomes. This supports evidence-based bias analysis. |
| 82 | B | SHAP provides post-hoc explanations for complex models, allowing organizations to retain predictive performance while satisfying explainability requirements. |
| 83 | B | Beneficence emphasizes improving human well-being while keeping humans involved in important decisions, especially in high-impact domains like healthcare. |
| 84 | B | Fairness metrics evaluate different properties. Satisfying one (e.g., Demographic Parity) does not imply satisfying another (e.g., Equalized Odds). |
| 85 | B | Representative data collection and stakeholder engagement during pre-development help identify population differences, such as regional dialects, before deployment. |
| 86 | B | Technical stakeholders require detailed implementation information, including training procedures, feature engineering, version history, and error analysis. |
| 87 | B | Model Cards communicate intended use, known limitations, appropriate deployment contexts, and risks to stakeholders. |
| 88 | B | Attendance may indirectly reveal disability status, making it a proxy feature that can introduce unintended discrimination. |
| 89 | B | Grad-CAM highlights image regions influencing CNN predictions and is commonly used for vision models. |
| 90 | A | Continuous monitoring with predefined retraining criteria is a hallmark of Responsible AI governance during post-deployment. |
| 91 | B | Local explanations describe why a specific prediction was made for an individual applicant. |

Q Answer Explanation

- 92 **B** Model Cards provide structured information that supports regulatory review, transparency, and governance.
- 93 **B** SHAP explains prediction contributions but does not determine whether a model is fair or unbiased.
- 94 **B** Post-processing is particularly useful when organizations cannot retrain proprietary or black-box models.
- 95 **B** Different decision thresholds across business units indicate inconsistent deployment governance and policy implementation.
- 96 **B** Explainability reveals which features influenced predictions, helping engineers detect spurious correlations and unintended model behavior.
- 97 **A** Fairness metrics often conflict; satisfying one fairness definition does not guarantee another.
- 98 **B** Responsible AI requires leadership to weigh documented benefits, risks, fairness impacts, and societal consequences before deployment.
- 99 **B** Transparency spans the entire AI lifecycle through comprehensive documentation, traceability, monitoring, and governance—not just explainable models.
- 100 **C** Module 4 emphasizes integrating ethics, fairness, transparency, explainability, documentation, accountability, and stakeholder communication throughout the AI development lifecycle.